

# NGUYEN Quang-Duy

0000-0002-3517-0945

(+33) 7 53 97 25 47

quang-duy.nguyen@cea.fr

<https://w3id.org/people/quang-duy.nguyen/>

"I may stutter in speaking, but not in philosophizing, inventing, and empathizing."



## Education

### 2017 – 2020 **PhD in Computer Science**

École Doctorale des Sciences Pour l'Ingénieur, University Clermont-Auvergne, France

UR TSCF, MathNUM département, INRAE (Irstea), Clermont-Ferrand, France

*Funding : Scholarship from the region Auvergne-Rhône-Alpes and FEDER*

*PhD defense : 16/07/2020*

### 2014 – 2015 **Master 2 in Computer Networking**

University Claude Bernard Lyon I, France

*Funding : Scholarship AUF-IFI ; Grant from IdEx of University of Strasbourg*

*Note : 15.31/20.00 - Rank : 1/60*

### 2013 – 2014 **Master 1 in Computer Science**

Institut de la Francophonie pour l'Informatique (IFI), Vietnam National University, Hanoi

*Funding : Scholarship AUF-IFI*

*Note : 14.20/20.00*

### 2007 – 2012 **Engineer of Information Technology in Computer Engineering**

Hanoi University of Science and Technology, Vietnam

*Note : 14.96/20.00*

## Experience

### 12/2021 – present **Postdoctoral Researcher → Research Scientist**

LSEA laboratory, DILS department, CEA List, Palaiseau, France

*Project : LocalSEA ; Papyrus4Manufacturing ; CoGeFlux ; OTPaaS ; F&3D DTs ; APS-INTEROP ; TARGET-X ; Ros6GBUSBridge ; RAASCEMAN ; DeCloud ; Cir4Fun ; CIRPASS-2*

*Keywords : Industry 4.0/5.0, AAS, Papyrus, Manufacturing, Supply Chain, MDE, ROS 1/2*

### 10/2020 – 09/2021 **Postdoctoral Researcher**

LTCI laboratory, INFRES department, Télécom Paris, Palaiseau

*Project : OPC UA PubSub for Caméléon*

*Keywords : IIoT, SNCF, OPC UA, Embedded System, Zephyr OS, Node-RED, C*

### 01/2017 – 07/2020 **PhD Student**

UR TSCF, MathNUM département, INRAE

*Project : ConnecSens ; AgroTechnoPôle*

*Keywords : Agriculture, IoT, Ontology, Reasoning, OWL, SWRL, Java, Python*

### 08/2016 – 10/2016 **Research Engineer**

Programmation, Networks and Systems team, LaBRI laboratory, Bordeaux, France

*Project : Le Palais de la Mémoire*

*Keywords : Sounds, Method of Loci, Beacons, BLE, Android, Java*

### 04/2015 – 07/2016 **Master 2 Intern → Research Engineer**

Networks team, ICube laboratory (CNRS), Strasbourg, France

*Project : FIT/IoT-lab*

*Keywords : Multihoming, Mobility, Contiki, TelosB, Python, C/C++*

## Scientific Publications\*

[PI.20] Carolyn Bernier, Fatima Danash, **Quang-Duy Nguyen**, Eva Blomqvist, Tarmo Robal, Riina Maigre, Michael Linck, Susanne Guth-Orlowski, Elio Hbeich and Theodor Chirvasuta, *"Ontology Diversity for Digital Product Passports : An Analysis"*, Proceedings of the 8th International Conference on Industry of the Future and Smart Manufacturing, Padua, Italy, 11/2026

[PI.19-JI.6] **Quang-Duy Nguyen**, Fatima Danash, Arnab Sinha, and Chokri Mraidha, *"Product Digital Twin Ontology : A Pivot for Bridging Asset Administration Shell-based and Ontology-based Digital Product Passports"*, Proceedings of the 22th European Conference on Modelling Foundations and Applications (ECMFA), Journal of Object Technology, 06/2026

[PI.18] **Quang-Duy Nguyen**, Guéréguin Der Sylvestre Sidibe, Marcos Didonet Del Fabro, and Fabien Balligand, *"Model-Driven Development of Data Space-Enabled Asset Administration Shell Digital Twins"*, Proceedings of the 23rd International Conference on Software Architecture Companion (ICSA-C), Amsterdam, Netherlands, 06/2026

[PI.17] **Quang-Duy Nguyen**, Kunal Suri, and Guéréguin Der Sylvestre Sidibe, *"Towards an LLM-Powered Expert System for AAS-based Product Digital Twin Development"*, Proceedings of the 23rd International Conference on Software Architecture Companion (ICSA-C), Amsterdam, Netherlands, 06/2026

[PI.16] **Quang-Duy Nguyen**, Kunal Suri, and Guéréguin Der Sylvestre Sidibe, *"Towards Engineering Product Digital Twins for Industry 5.0 : Definition and Modeling Approach"*, Proceedings of the 30th International Conference on Emerging Technologies and Factory Automation (ETFA), Porto, Portugal, 09/2025

[PI.15] **Quang-Duy Nguyen**, Darine Rammal, Christophe Gaston, Deepak V Katkoria, Arnault Lapitre, and Saadia Dhouib, *"Towards Bridging Industrial Ethernet Networks : Protocol Translation and Runtime Verification"*, Proceedings of the 30th International Conference on Emerging Technologies and Factory Automation (ETFA), Porto, Portugal, 09/2025

[PF.3] **Quang-Duy Nguyen**, and Guéréguin Der Sylvestre Sidibe, *"Appliquer les jumeaux numériques industriels à l'agriculture : une perspective pour le déploiement des systèmes contextuels"*, La 8ème édition de l'Atelier INTégration de sources/masses de données hétérogènes et Ontologies, dans le domaine des sciences du VIVant et de l'Environnement (IN-OVIVE), adossé à la conférence Ingénierie des Connaissances (IC), Dijon, France, 07/2025

[PI.14-JI.5] **Quang-Duy Nguyen**, Maxime Morin, Saadia Dhouib, and Nicolas Navarro, *"An AAS-based Architecture for Plug and Produce and Order-Driven Production"*, Proceedings of the 11th IFAC Conference on Manufacturing Modelling, Management and Control (MIM), IFAC-PapersOnline, Trondheim, Norway, 06/2025

[PI.13] **Quang-Duy Nguyen**, and Saadia Dhouib, *"Lessons from Developing AAS Digital Twins for Factory Buildings with Papyrus4Manufacturing"*, Proceedings of the 3rd IEEE Industrial Electronics Society Annual Online Conference (ONCON), Beijing, China, 12/2024

[PI.12] **Quang-Duy Nguyen**, Yining Huang, Guéréguin Der Sylvestre Sidibe, and Saadia Dhouib, *"From Multiple Digital Twins to a Multi-Faceted Digital Twin : Towards an AAS-Based Approach"*, Proceedings of the 29th International Conference on Emerging Technologies and Factory Automation (ETFA), Padova, Italy, 09/2024

[PI.11] **Quang-Duy Nguyen**, Saadia Dhouib, Eric Lucet, Antoine Le Mortellec, and Fabien Baligand, *"Bridging the Gap between IT and OT with AAS Digital Twins and MDE Techniques : An Industrial Waste Management Case Study"*, Proceedings of the 29th International Conference on Emerging Technologies and Factory Automation (ETFA), Padova, Italy, 09/2024

[JI.4] **Quang-Duy Nguyen**, Yining Huang, François Keith, Christophe Leroy, Minh-Thuyen Thi, and Saadia Dhouib, *"Manufacturing 4.0 : Checking the Feasibility of a Work Cell Using Asset Administration Shell and Physics-Based Three-Dimensional Digital Twins"*, MDPI Machines, 01/2024

### Summary

First-authored items : 22/27

ISI-indexed Items : 14

Scopus-indexed Items : 21

Citations :

GS : 136, RG : 123,

Scopus : 99, WoS : 53

H-index by Scopus : 6

Items classified by SJR :

Q1 : 1, Q2 : 2,

Q3 : 1, Q4 : 2

Items classified by domains :

IoT : 9, OPC UA : 8,

Ontology : 8, AAS DT : 11

- [PI.10-JI.3] Fadwa Rekik, Saadia Dhouib, and **Quang-Duy Nguyen**, *"Bridging the Gap between SysML and OPC UA Information Models for Industry 4.0"*, Proceedings of the 19th European Conference on Modelling Foundations and Applications (ECMFA), Journal of Object Technology, 07/2023
- [PI.9] **Quang-Duy Nguyen**, Saadia Dhouib, Yining Huang, and Patrick Bellot, *"An Approach to Bridge ROS 1 and ROS 2 Devices into an OPC UA-based Testbed for Industry 4.0"*, Proceedings of the 1st IEEE Industrial Electronics Society Annual Online Conference (ONCON), Kharagpur, India, 12/2022
- [PI.8] **Quang-Duy Nguyen**, Saadia Dhouib, and Patrick Bellot, *"A Unified Method to Design Bridges for OPC UA PubSub Networks in the Industrial IoT"*, Proceedings of the 27th IEEE International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD), Paris, France, 11/2022
- [PI.7] **Quang-Duy Nguyen**, Saadia Dhouib, Kunal Suri, and Fadwa Tmar, *"From Requirement Specification to OPC UA Information Model Design : A Product Assembly Line Monitoring Case Study"*, Proceedings of the 20th IEEE International Conference on Industrial Informatics (INDIN), Perth, Australia, 07/2022
- [PI.6] **Quang-Duy Nguyen**, Patrick Bellot, and Pierre-Yves Petton, *"An OPC UA PubSub Implementation Approach for Memory-Constrained Sensor Devices"*, Proceedings of the 31st IEEE International Symposium on Industrial Electronics (ISIE), Anchorage, United States, 06/2022
- [PI.5] **Quang-Duy Nguyen**, Fadwa Tmar, Yining Huang, and Saadia Dhouib, *"Early Lessons Learned from the Development of a Local OPC UA-based Robotic Testbed for Research"*, Proceedings of the 31st IEEE International Symposium on Industrial Electronics (ISIE), Anchorage, United States, 06/2022
- [PI.4] **Quang-Duy Nguyen**, Saadia Dhouib, Jean-Pierre Chanet, and Patrick Bellot, *"Towards a Web-of-Things Approach for OPC UA Field Device Discovery in the Industrial IoT"*, Proceedings of the 18th IEEE International Conference on Factory Communication Systems (WFCS), Pavia, Italy, 04/2022
- [PI.3] **Quang-Duy Nguyen**, Catherine Roussey, Patrick Bellot, and Jean-Pierre Chanet, *"Stack of Services for Context-Aware Systems : An Internet-Of-Things System Design Approach"*, Proceedings of the 15th IEEE International Conference on Computing and Communication Technologies (RIVF), Hanoi, Vietnam, 08/2021
- [T.1] **Quang-Duy Nguyen**, *"Interoperability and Upgradability Improvement for Context-Aware Systems in Agriculture 4.0"*, PhD thesis, University Clermont-Auvergne, France, 2020
- [JI.2] Julian Eduardo Plazas, Sandro Bimonte, Christophe de Vault, Michel Schneider, **Quang-Duy Nguyen**, Jean-Pierre Chanet, Hongling Shi, Kun Mean Hou, and Juan Carlos Corrales, *"A Conceptual Data Model and its Automatic Implementation for IoT-Based BI Applications : UML Profile and MDA Approach"*, IEEE Internet of Things Journal, 10/2020
- [PF.2] **Quang-Duy Nguyen**, Catherine Roussey, Maria Poveda-Villalón, Christophe de Vault, Jean-Pierre Chanet, and Camille Noûs, *"CASO et IRRIG deux ontologies pour le développement de systèmes contextuels : cas d'usage sur l'automatisation de l'irrigation"*, Actes des 31es journées francophones d'Ingénierie des Connaissances (IC), Angers, France, 07/2020
- [JI.1] **Quang-Duy Nguyen**, Catherine Roussey, Maria Poveda-Villalón, Christophe de Vault, and Jean-Pierre Chanet, *"Development Experience of a Context-Aware System for Smart Irrigation using CASO and IRRIG ontologies"*, Special Issue "Semantic Technologies Applied to Agriculture", MDPI Applied Sciences, 03/2020
- [PI.2] Maria Poveda-Villalón, **Quang-Duy Nguyen**, Catherine Roussey, Christophe de Vault, and Jean-Pierre Chanet, *"Ontological Requirement Specification for Smart Irrigation Systems : A SOSA/SSN and SAREF Comparison"*, Proceedings of the 9th International Semantic Sensor Networks Workshop (SSN) of the 17th International Semantic Web Conference (ISWC), Monterey, United States, 10/2018
- [PF.1] Maria Poveda-Villalón, **Quang-Duy Nguyen**, Catherine Roussey, Christophe de Vault, and Jean-Pierre Chanet, *"Besoins ontologiques d'un système d'irrigation intelligent : Comparaison entre SSN et SAREF"*, Actes des 29es journées francophones d'Ingénierie des Connaissances (IC), Nancy, France, 07/2018
- [PI.1] **Quang-Duy Nguyen**, Julien Montavont, Nicolas Montavont, and Thomas Noël, *"RPL Border Router Redundancy in the Internet of Things"*, Proceedings of the 15th International Conference on Ad-Hoc Networks and Wireless (ADHOC-NOW), Lille, France, 07/2016

## Intellectual Property

1. Patent : System and Method for Automatic Digital Twins Deployment  
Status : Pending
2. Patent : System and Method for Runtime Configuration of a Network Protocol Bridge Device  
Status : Pending
3. Patent : E-Tickets Management System and Method  
Status : Pending
4. Software Deposit : Papyrus4Manufacturing 2025  
Status : Registered      Number : IDDN.FR.001.030014.000.S.C.2026.000.10000

## Peer Review Activities

2025 1 x Review for **International Journal of Critical Infrastructure Protection** (IJCIP), Elsevier  
2025 1 x Review for **Machines**, MDPI  
2025 2 x Reviews for **Advanced Engineering Informatics** (AEI), Elsevier  
2026 1 x Review for **Computer Networks** (COMNET), Elsevier

## Research Skills

|                       |                     |
|-----------------------|---------------------|
| Project Management    | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| System Implementation | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| Scientific Writing    | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| Presentation          | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |

## Linguistic

|            |                     |
|------------|---------------------|
| Vietnamese | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| French     | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| English    | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |

## Technical Skills

|               |                                       |
|---------------|---------------------------------------|
| Standard      | OPC UA, AAS, RAMI 4.0, ISA-95         |
| IT Network    | TCP/IP, MQTT, REST, WebSocket         |
| OT Network    | UA FX, EtherNET/IP, MODBUS, CAN       |
| Cybersecurity | AES-CTR, TLS, JWT                     |
| OS/Middleware | Linux, Zephyr, BaSyx, ROS2            |
| Methodology   | Mini-Waterfall, LOT, Agile, MDE       |
| Vocabulary    | SOSA/SSN, CASO, IRRIG, PDTO           |
| Tool          | Papyrus, Protégé, Wireshark, UaExpert |

## Programming

|              |                     |
|--------------|---------------------|
| C            | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| Java         | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| C++          | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| Python       | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| Shell script | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| Node-Red/JS  | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |
| SQL/SPARQL   | ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ |